



Tennessee Valley Authority, Sequoyah Nuclear Plant, P.O. Box 2000, Soddy Daisy, Tennessee 37384

February 13, 2020

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Sequoyah Nuclear Plant, Units 1 and 2
Renewed Facility Operating License No. DPR-77 and DPR-79
NRC Docket Nos. 50-327 and 50-328

Subject: **Licensee Event Report 50-327 and 50-328/2019-004-00, Automatic Actuation of Emergency Diesel Generators Due to Loss of Power to 6.9kV Shutdown Board**

The enclosed licensee event report provides details concerning an automatic actuation of the four emergency diesel generators as a result of the loss of power to a 6.9kV shutdown board. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(iv)(A), as any event or condition that resulted in manual or automatic actuation of emergency AC electrical power systems, including emergency diesel generators.

There are no regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact Mr. Jeffrey Sowa, Site Licensing Manager, at (423) 843-8129.

Respectfully,

A handwritten signature in black ink, appearing to be "MR", with a long horizontal flourish extending to the right.

Matthew Rasmussen
Site Vice President
Sequoyah Nuclear Plant

Enclosure: Licensee Event Report 50-327/2019-004-00

cc: NRC Regional Administrator – Region II
NRC Senior Resident Inspector – Sequoyah Nuclear Plant



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

Sequoyah Nuclear Plant Unit 1

2. Docket Number

05000327

3. Page

1 OF 6

4. Title

Automatic Actuation of Emergency Diesel Generators Due to Loss of Power to 6.9KV Shutdown Board

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
12	16	2019	2019	004	00	02	13	2020	Sequoyah Nuclear Plant Unit 2	05000328
									Facility Name	Docket Number
									NA	05000

9. Operating Mode 1	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)

10. Power Level 100	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(iii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact

Andrew McNeil

Telephone Number (Include Area Code)

423-843-8098

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To ICES	Cause	System	Component	Manufacturer	Reportable To ICES
B	EB	51	B455	Y					

14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

15. Expected Submission Date

Month

Day

Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On December 16, 2019, at 03:58 Eastern Standard Time (EST), Sequoyah Nuclear Plant Unit 1 experienced a loss of 1B-B 6.9 kilo-Volt (KV) Shutdown Board offsite power when the normal feeder breaker from the 1C 6.9 KV Unit Board to the 1B-B 6.9 KV Shutdown Board tripped. This resulted in a loss of offsite power to the 1B-B 6.9 KV Shutdown Board, which automatically started all four emergency diesel generators as designed. The 1B-B 6.9 KV Shutdown Board was automatically reenergized from its associated emergency diesel generator, and was subsequently transferred back to an alternate offsite power supply.

The cause of the feeder breaker trip was an overcurrent relay component failure, specifically a metal oxide varistor of the incorrect voltage rating, which actuated the breaker overcurrent protection. The failed relay was replaced with a new relay of the proper voltage rating. An extent of condition will be performed as plant conditions permit to evaluate any other 250 VDC rated applications that may have incorrect voltage components.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Sequoyah Nuclear Plant Unit 1	05000-327	YEAR 2019	SEQUENTIAL NUMBER - 004	REV NO. - 00

NARRATIVE**I. Plant Operating Conditions Before the Event**

Sequoyah Nuclear Plant (SQN) Unit 1 was at 100 percent Rated Thermal Power (RTP). SQN Unit 2 was at 47 percent RTP conducting start-up and power ascension activities following a reactor trip on December 12, 2019.

II. Description of Event**A. Event Summary:**

On December 16, 2019, at 0358 Eastern Standard Time (EST), SQN Unit 1 experienced a loss of 1B-B 6.9 kilo-Volt (KV) Shutdown Board (SDBD) [EIS: ECBD] offsite power when the non-safety related normal feeder breaker [EIS: BKR, 52] from the 1C 6.9 KV Unit Board to the 1B-B 6.9 KV SDBD tripped. Loss of offsite power to the 1B-B SDBD automatically started all four emergency diesel generators (EDGs) [EIS: DG] as designed.

The 1B-B KV SDBD was automatically energized from the 1B-B 6.9 KV Diesel Generator. All required 6.9 KV loads were sequenced back on to the 1B-B 6.9 KV SDBD as designed after the board was reenergized. The rest of the electrical system remained in a normal alignment.

A Preventative Maintenance (PM) activity on 250 Volts-Direct Current (VDC) Station Battery #1 [EIS: BTRY] was in progress at the time of the event. During the performance of the PM activity, 250 VDC Station Battery #1 was placed on an equalizing charge with the battery in service. Upon initiation of the battery charge, the voltage rose to approximately 300 VDC and was adjusted down to 280 VDC.

The 300 VDC experienced on the 250 VDC system [EIS: EI] was high enough to damage the installed Asea Brown Boveri (ABB) GR-5 Relay used in the 1722 Breaker 51G [EIS: 51] location, causing the breaker to trip open. During post-transient investigation, it was discovered that although the relay was marked appropriately as a 250 VDC model, it had the wrong voltage rated metal oxide varistor (described from this point as "varistor") installed. The installed varistors were 150 VDC rated components and were original to the relay from manufacturing, prior to its installation in 2014.

An 8-hour non-emergency event notification (EN 54446) was completed on December 16, 2019 at 0912 EST. This event is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as any event or condition that resulted in manual or automatic actuation of emergency AC electrical power systems, including EDGs.

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- B. Status of structures, components, or systems that were inoperable at the start of the event and contributed to the event:

No inoperable structures, components, or systems contributed to this event.

- C. Dates and approximate times of occurrences:

Date/Time (EST)	Description
March 3, 2014	Defective ABB GR-5 Relay installed
December 16, 2019 at 0355	Approximate time of multiple alarms received in Main Control Room indicating a possible electrical malfunction.
December 16, 2019 at 0356	Unit 1 1B-B 6.9 KV SDBD experiences a loss-of-offsite power, triggering Engineered Safety Features (ESF) and starting EDGs. 1B-B 6.9 KV SDBD automatically energized from the 1B-B DG.
December 16, 2019 at 0358	Main Control Room Operators entered into an abnormal operating procedure for partial loss of offsite power.
December 16, 2019 at 0835	1B-B DG disconnected from 1B-B 6.9 KV SDBD, with offsite power restored.
December 16, 2019 at 1458	All EDGs secured and restored to standby alignment.
December 17, 2019	Defective ABB GR-5 Relay replaced.

- D. Manufacturer and model number of each component that failed during the event:

Littelfuse ABB GR-5 Metal Oxide Varistor

- E. Other systems or secondary functions affected:

There were no systems or secondary functions affected by this event.

- F. Method of discovery of each component or system failure or procedural error:

Alarms were initially received in the Main Control Room indicating an electrical distribution related event, triggering entry into the Abnormal Operating Procedure (AOP).

- G. Failure mode, mechanism, and effect of each failed component:

The varistors provide the implementation signal for the 250 VDC 51G Overcurrent Relay protection for the affected breaker. When the varistors were subjected to a voltage beyond their 150 VDC design capability, they failed, providing signal to the 51G relay to open the affected breaker. The opening of the breaker de-energized the safety-related electrical

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board, generating an undervoltage signal which started the EDGs.

H. Operator actions:

Operators appropriately responded to the event by utilizing alarm response procedures and the AOP.

I. Automatically and manually initiated safety system responses:

The opening of the unit board breaker de-energized the safety-related electrical board, generating an undervoltage signal which started the EDGs.

III. Cause of the Event

A. Cause of each component or system failure or personnel error:

Inspection of the failed relay control board revealed improperly rated metal oxide varistors. Relay function 51G is implemented with an ABB GR-5 solid state relay supplied "commercial grade". The GR-5 circuit board involved in the event was found to have failed catastrophically. The failed circuit board was found to have varistors installed with a manufacturer specified maximum continuous voltage of 200 VDC. The correct varistors for the application required a manufacturer specified maximum continuous voltage of 330 VDC.

B. Cause(s) and circumstances for each human performance related root cause:

There was no identified human performance related root cause.

IV. Analysis of the Event:

The Class 1E alternating current (AC) Electrical Power Distribution System [EIS: EA] sources consist of the offsite power sources (preferred power sources), and the onsite standby power sources (Train A and Train B EDGs). The design of the AC electrical power system provides independence and redundancy to ensure an available source of power to the Engineered Safety Feature (ESF) systems.

The onsite Class 1E AC Electrical Power Distribution System is divided into two redundant and independent load groups with two 6.9 KV SDBDs in each load group. Each 6.9 KV SDBD has a connection to a preferred offsite power source and an EDG. The 6.9 KV SDBDs in a load group (i.e., 1A-A and 2A-A, or 1B-B and 2B-B) are normally powered by the same offsite power circuit. Each 6.9 KV SDBD can also be powered by a dedicated EDG. Two EDGs associated with one load group can provide all safety related functions to mitigate a loss of coolant accident (LOCA) in one unit and safely shut down the other unit. The Train A and Train B ESF systems each provide

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for the minimum safety functions necessary to shut down the plant and maintain it in a safe shutdown condition.

An EDG starts automatically on a safety injection (SI) signal or on a 6.9 KV SDBD degraded voltage or loss-of-voltage signal. After the EDG has started, it will automatically tie to its respective 6.9 KV SDBD after offsite power is tripped as a consequence of a 6.9 KV SDBD degraded voltage or loss-of-voltage signal, independent of or coincident with an SI signal. Following the trip of offsite power, a loss-of-voltage signal strips nonpermanent loads from the 6.9 KV SDBD. When the EDG is tied to the 6.9 KV SDBD, loads are then sequentially connected to its respective 6.9 KV SDBD by individual load sequence timers.

In this event, the 1B-B 6.9 KV SDBD was immediately reenergized by its associated EDG. Throughout the transient, both trains of the ESF system were available as designed to provide for the minimum safety functions necessary to shut down both units and maintain them in a safe shutdown condition.

V. Assessment of Safety Consequences

There were no actual safety consequences as a result of this event.

- A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event:

The alternate offsite AC source of power remained available to the 1B-B 6.9 KV SDBD during 1B-B EDG operation.

- B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident:

The event did not occur when the reactor was shut down.

- C. For failure that rendered a train of a safety system inoperable, an estimate of the elapsed time from discovery of the failure until the train was returned to service:

The duration of the loss of offsite power to the 1B-B SDBD was 4 hours 39 minutes.

VI. Corrective Actions

The event was entered into the Tennessee Valley Authority Corrective Action Program (CAP) under Condition Report #1573472 and 1573726.

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A. Immediate Corrective Actions:

The failed GR-5 relay was replaced with a new relay with the proper voltage rated varistors installed.

B. Corrective Actions to Prevent Recurrence or to reduce probability of similar events occurring in the future:

An extent of condition will be performed as plant conditions permit to evaluate any other 250 VDC rated applications that may have incorrect voltage components.

VII. Previous Similar Events at the Same Site:

There were no previous similar events at SQN occurring within the last three years.

VIII. Additional Information

There is no additional information.

IX. Commitments:

There are no commitments.